Smart Passenger Detection System (SPDS)

Nouf Ibrahim AlDawsari – 44209442

Albatul Abdullah Albogami – 44205933

Supervised by:

Dr. Manahill Idriss Adrob Anja

Department of Computer Science

Collage of Computers and Information Technology

Taif University, KSA

2025-2026

Contact Information

Authors Information		
Name	**Number**	**Email**
Nouf Ibrahim AlDawsari	**0541262715**	**S44209442@students.tu.edu.sa**
Albatul Abdullah Albogami	**0507059207**	**S44205933@students**

Supervised

Name: Dr. Manahill Idriss Adrob Anja

Department of Computer Science

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Authors:

Name: Albatul Abdullah	**Signature:____________**	**Date: 14-9-2025**
Name: Nouf Ibrahim	**Signature:____________**	**Date: 14-9-2025**

Supervisor:

Name: Dr. Manahill Idriss Adrob Anja	**Signature:** Manahill	**Date: 14-9-2025**

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Authors:

Name: Albatul Abdullah	**Signature:__________**	**Date: 14-9-2025**
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Dedication

This project is lovingly dedicated to our beloved parents,
whose endless support, prayers, and encouragement have been our guiding light throughout this journey.

To our families, for their patience and motivation during the challenging times.

To our supervisor, **Dr. Manahill Idriss Adrob Anja**,
for her continuous guidance, valuable feedback, and inspiration that helped us complete this work.

And to everyone who believed in us and supported our vision of creating a safer environment for children
this achievement is for you.

Abstract

School bus safety remains a critical concern, particularly in cases where children are accidentally left inside buses after trips. Existing solutions often rely on manual inspection, which is prone to human error, or on camera-based systems that raise privacy and cost concerns. Furthermore, many existing approaches are not well adapted to the environmental and operational conditions in Saudi Arabia.

The Smart Passenger Detection System (SPDS) was developed to address this problem by automatically detecting seat occupancy after the engine was turned off. The system alerted the driver through audio-visual signals, reduced the likelihood of human inspection errors, and provided a low-cost and scalable solution suitable for school-bus safety applications.

The methodology of the project included reviewing related studies, identifying research gaps, designing the system architecture and use cases, selecting the required hardware and software components, implementing the System, and testing its operation under different conditions.

The results showed that pressure-based sensing, combined with simple alert mechanisms, offered a reliable and privacy-preserving solution for passenger detection. The system improved school-bus safety by providing immediate feedback to the driver and reducing the risk of preventable accidents.

Keywords:

Smart bus safety, passenger detection, pressure sensors, Arduino-based systems, child protection.

ملخص مقترح البحث (عربي)

تُعد سلامة الحافلات المدرسية من القضايا الحساسة، خاصة في الحالات التي يُترك فيها الأطفال داخل الحافلات بعد انتهاء الرحلات. وتعتمد الحلول الحالية في الغالب على التفتيش اليدوي، وهو أسلوب معرّض للخطأ البشري، أو على أنظمة تعتمد على الكاميرات، وهي أنظمة مرتفعة التكلفة وتثير مخاوف تتعلق بالخصوصية. بالإضافة إلى ذلك، فإن كثيرًا من الحلول المتوفرة لا تتكيف بصورة كافية مع الظروف البيئية والتشغيلية في المملكة العربية السعودية.

تم تطوير نظام الكشف الذكي للركاب (SPDS) لمعالجة هذه المشكلة من خلال الكشف التلقائي عن وجود راكب على المقعد بعد إطفاء المحرك. وقد وفر النظام المنفذ تنبيهًا فوريًا للسائق باستخدام إشارات صوتية ومرئية، وساهم في تقليل احتمالية الخطأ البشري المرتبط بالتفتيش اليدوي، كما قدم حلًا منخفض التكلفة وقابلًا للتوسع ومناسبًا لتطبيقات سلامة الحافلات المدرسية.

اعتمدت منهجية المشروع على مراجعة الدراسات السابقة، وتحديد الفجوات البحثية، وتصميم بنية النظام وحالات الاستخدام، واختيار المكونات البرمجية والعتادية المناسبة، ثم تنفيذ النموذج الأولي واختبار أدائه تحت ظروف تشغيل مختلفة.

وأظهرت النتائج أن الاعتماد على حساسات الضغط، إلى جانب آليات التنبيه البسيطة مثل شاشة LCD والجرس والضوء التحذيري، يوفر حلًا عمليًا وموثوقًا ومراعيًا للخصوصية للكشف عن الركاب. كما أسهم النظام في تعزيز سلامة الحافلات المدرسية من خلال توفير تنبيه فوري للسائق وتقليل خطر الحوادث التي يمكن الوقاية منها.

الكلمات المفتاحية:

سلامة الحافلات الذكية، كشف الركاب، حساسات الضغط، الأنظمة المعتمدة على الأردوينو، حماية الأطفال.

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Introduction

1.1 Overview and Problem Background

School buses have recently been considered one of the most important transportation safety concerns worldwide because of incidents in which children were accidentally left alone inside school buses. Such accidents are often linked to human carelessness and the lack of reliable automated safety systems. Although the consequences of these incidents are severe, school transportation systems still rely heavily on manual inspection by drivers or supervisors, which makes them vulnerable to human error, fatigue, and oversight. Incidents reported in Saudi Arabia have highlighted the seriousness of this problem and the need for effective preventive solutions.

When a child or passenger is left inside a closed vehicle after the engine is turned off, especially in hot weather, the internal temperature may rise rapidly to dangerous and even fatal levels. According to the U.S. National Highway Traffic Safety Administration (NHTSA), a child's body temperature rises three to five times faster than an adult's, heatstroke begins when the core body temperature reaches about 104°F, and a core temperature of about 107°F can be lethal. These facts explain why even a short delay in detecting a child left inside a bus can lead to tragedy.

Traditional safety procedures, including visual inspection and manual attendance checks, cannot always guarantee that all passengers have left the vehicle. A sleeping, silent, or unnoticed child may easily be missed, particularly during rushed schedules or crowded trips. This repeated risk clearly shows the need for an intelligent system that can automatically detect the presence of passengers inside the bus after the engine is turned off and provide immediate alerts.

This is the gap addressed by the proposed Smart Passenger Detection System (SPDS), which uses seat pressure sensors, an Arduino-based controller, and audio-visual alert devices. In the implemented system, seat occupancy was monitored after the engine state changed to OFF, and an alarm was activated if a passenger was still detected. The system provided a low-cost, scalable, and privacy-preserving solution intended to improve safety, reduce human error, and help prevent avoidable tragedies.

In addition, this project aligns with Saudi Vision 2030, which emphasizes improving quality of life and enhancing traffic safety through practical technological solutions.

1.2 Problem Statement

One of the major challenges in school and public transportation safety is ensuring that buses are completely empty of passengers when they are turned off or parked. Drivers and supervisors may overlook a sleeping child or fail to perform a proper final inspection. This negligence creates a serious health risk, as children left inside closed buses may be exposed to rapidly rising temperatures, dehydration, suffocation, heatstroke, and even death within a short period of time.

Therefore, the problem lies not only in human error, but also in the absence of a reliable technological safeguard that guarantees final verification of passenger evacuation. Without such a safeguard, the likelihood of repeated tragedies remains high.

This situation presents several key challenges:

- Difficulty in Ensuring Evacuation: buses may not be thoroughly checked when the engine is turned off or the vehicle is parked.
- Human Error Risk: drivers or supervisors may overlook sleeping children or fail to perform a complete inspection.
- Serious Health Risks Caused by Negligence: children left inside closed buses may face heatstroke, dehydration, suffocation, cardiac arrest, and possible death within a short period.
- Lack of Reliable Safeguards: the absence of a smart technological system to verify passenger evacuation increases the likelihood of repeated tragedies.

This project addressed these challenges by proposing a reliable technological safeguard that helps ensure buses are thoroughly checked and fully evacuated, thereby protecting children's lives and preventing avoidable tragedies.

1.3 Project Objectives

The main objectives of this project are:

- To enhance passenger safety by detecting weight/pressure on bus seats after the engine is turned off.
- To prevent incidents of children being forgotten inside school buses through automatic alerts.
- To provide instant alerts using audio and visual indicators.
- To develop a simple and low-cost system using Arduino, seat pressure sensors, a buzzer, an LED, an LCD screen, and control buttons.
- To support smart and safe transportation within school environments and reduce human error.

1.4 Scope

The project focuses on improving safety inside school and passenger buses by detecting any remaining passengers after the trip. The system uses seat pressure sensors connected to an Arduino controller to identify whether any seat remains occupied after the engine is turned off.

When occupancy is detected, the system automatically activates sound and light alerts to notify the driver or supervisor. The current version of the project is limited to local detection and alerts. However, it can be expanded in the future to include smart features such as mobile notifications or IoT connectivity.

1.5 Contributions / Significance of the Project

- Improves safety in school and passenger buses by detecting seat occupancy after trips.
- Provides a low-cost and easy-to-install system using Arduino and seat pressure sensors.
- Prevents accidents caused by forgetting children or passengers inside buses.
- Reduces human error through automated detection and alerts.
- Supports Saudi Arabia's Vision 2030 for smart and safe transportation.
- It can be expanded in the future to include mobile notifications or IoT connectivity.

1.6 Activity Plan

Figure 1.1 presents the activity plan followed in developing the Smart Passenger Detection System (SPDS), including problem definition, design, implementation, testing, evaluation, and report preparation

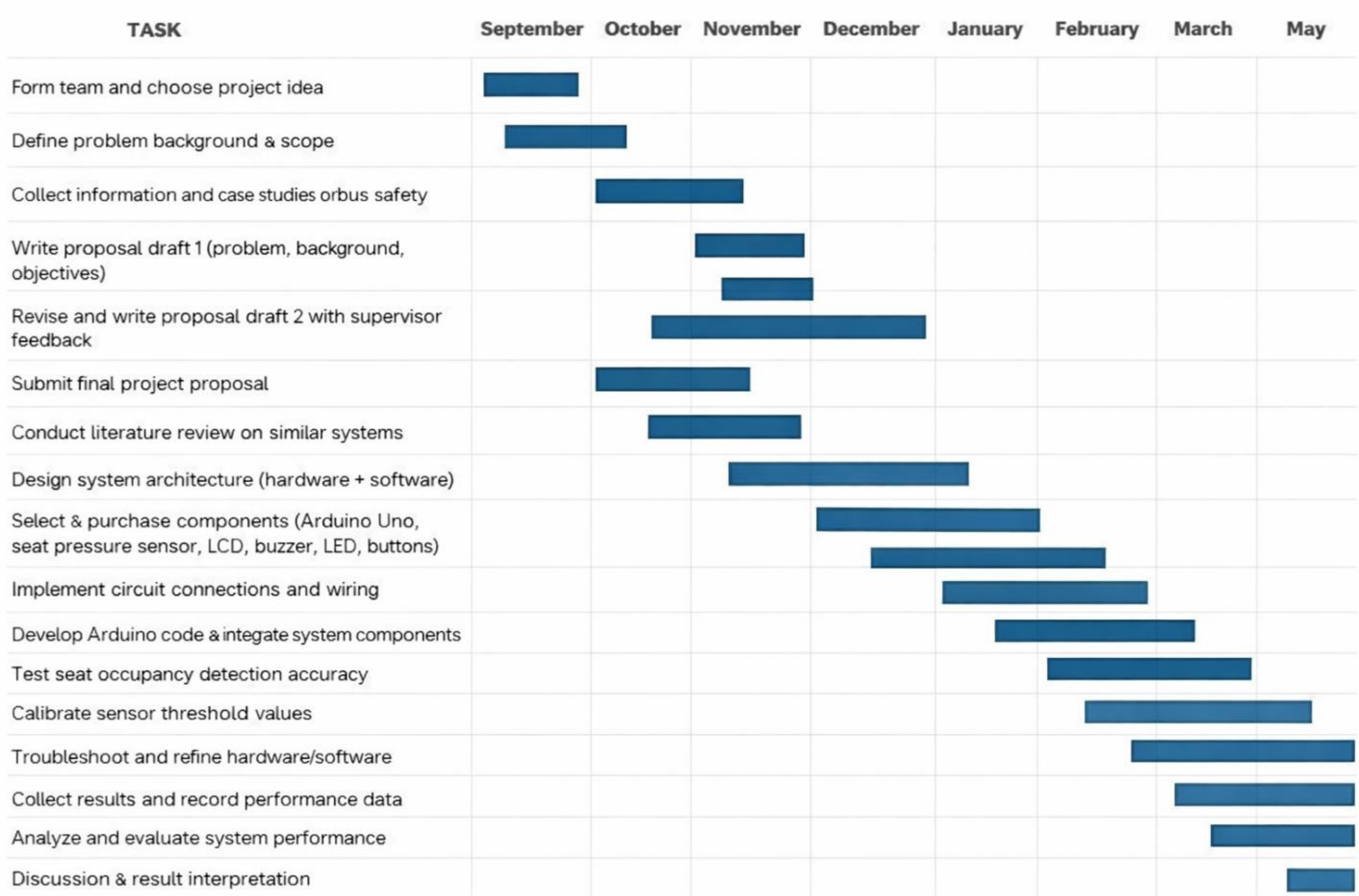

Figure 1: Activity Plan

1.7 Summary of the Remaining Chapters

This chapter introduced the Smart Bus Passenger Detection System (SPDS), outlining the background, problem, objectives, scope, and significance of the project. It addressed the critical issue of children being forgotten inside school buses and emphasized the need for a simple, automated solution that relies on seat-pressure detection. The proposed system uses reliable and low-cost components—such as force-sensitive seat sensors, audio-visual alerts, and an Arduino microcontroller—to ensure timely detection and improve transportation safety.

Chapter 2 – Literature Review: This chapter reviews previous studies and existing systems related to bus safety and passenger detection. It identifies the limitations of current methods and justifies the need for a more efficient and low-cost smart solution.

Chapter 3 – System Design and Implementation: This chapter explains the design and technical structure of the Smart Bus Passenger Detection System (SPDS), including hardware components, circuit diagrams, and Arduino programming. It also describes how the seat pressure interacts with the alarm system to ensure safety.

Chapter 4 – Results and Discussion: This chapter presents the results of testing and implementation. It evaluates the system's accuracy, performance, and effectiveness in detecting passengers under different conditions. Observations and improvements made during the testing phase are also discussed.

Chapter 5 – Conclusion and Future Work: The final chapter summarizes the project's key outcomes and assesses how well the objectives were achieved. It also proposes possible future enhancements, such as adding mobile notifications or integrating IoT features for remote monitoring and data collection.

1.8 Conclusion

This chapter provided an overview of the Smart Bus Safety System, explaining the background of the problem, its causes, and its impact on transportation safety. It identified the main challenge of ensuring that buses are completely empty after each trip and highlighted the need for a smart system to automatically detect any remaining passengers. The chapter also defined the project's objectives, scope, and significance, showing how the proposed system can reduce human error and prevent tragic accidents inside school buses. The next chapter, Chapter 2: Literature Review, will discuss previous studies and existing systems related to bus safety and seat-pressure detection. It will compare different approaches and explain how the proposed system improves upon existing solutions through its simplicity, affordability, and practical application.

Literature Review

2.1 Introduction

In recent years, transportation safety has become an important area of research in both academia and industry. Increasing attention has been given to smart monitoring technologies that can reduce accidents, improve operational safety, and prevent passengers from being left inside vehicles after the end of a trip. Traditional safety procedures in school and public buses still depend heavily on manual inspection by drivers or supervisors, which may fail because of fatigue, time pressure, or human error.

As a result, automated passenger detection and monitoring systems have become increasingly important for improving bus safety. Different approaches have been proposed in previous studies, including computer vision, Internet of Things (IoT) monitoring, wireless sensor networks, and seat-occupancy sensing. Each approach offers advantages, but also presents limitations related to cost, complexity, privacy, or environmental sensitivity.

This chapter reviews and analyzes several related studies on passenger detection, seat-occupancy monitoring, and passenger-counting systems. The aim is to examine their methodologies, strengths, and limitations, and then compare them with the proposed Smart Passenger Detection System (SPDS). This comparison helps identify the research gaps addressed by SPDS and justifies the design choices adopted in this project.

2.2 Summary of Related Works

2.2.1 Study A: Passenger Counting via Deep Learning

Reference: Estimation of the Number of Passengers in a Bus Using Deep Learning (MDPI, 2020)

This study proposed a deep-learning approach for estimating the number of passengers inside buses using front- and rear-mounted cameras. It employed YOLOv3 and autoencoder-based models to detect and estimate passenger presence under different operating conditions. The study reported good counting performance and demonstrated that computer vision can provide high detection accuracy in controlled environments.

Strengths:

- High detection accuracy.
- Full visual coverage of the bus interior.
- Useful for large-scale counting applications.

Weaknesses:

- Sensitive to lighting conditions and camera quality.
- Higher hardware and processing cost.
- Raises privacy concerns because it depends on continuous video capture.

Comparison with SPDS:

Unlike camera-based deep-learning systems, SPDS uses pressure-based sensing and avoids the use of cameras entirely. This makes SPDS more privacy-preserving, simpler, and less costly, especially for school-bus safety applications where the main goal is detecting whether a passenger remains in the bus after the engine is turned off rather than counting all passengers.

2.2.2 Study B: IoT-Based Seat Occupancy Monitoring System

Reference: Smart Occupancy Monitoring System Using IoT Technologies (SSRN, 2023)

This study introduced a seat-occupancy monitoring system using Force Sensitive Resistor (FSR) sensors integrated with an Arduino Uno and IoT modules. The purpose of the system was to monitor seat usage and help prevent bus overloading through real-time data transmission.

Strengths:

- Low-cost and practical implementation.
- Scalable seat-based detection approach.
- Suitable for occupancy monitoring applications.

Weaknesses:

- Focused mainly on seat occupancy rather than post-trip safety.
- Limited to seated passengers only.
- Added communication complexity through IoT integration.

Comparison with SPDS:

SPDS builds on the same seat-pressure concept but focuses specifically on post-trip passenger detection rather than continuous occupancy monitoring. In the current System, SPDS uses local audio-visual alerts instead of IoT connectivity, which simplifies the system and makes it more practical for a low-cost school-bus safety solution.

2.2.3 Study C: Review of Passenger Counting Technologies

Reference: A Review of Passenger Counting in Public Transport Concepts with Solution Proposal (MDPI, 2022)

This review classified several passenger-counting technologies, including ticketing systems, ultrasonic sensors, Wi-Fi-based detection, LiDAR, and camera-based methods. It compared these technologies in terms of cost, scalability, accuracy, and privacy.

Strengths:

- Provides a broad classification of available technologies.
- Useful for selecting an appropriate sensing approach.

- Highlights trade-offs between accuracy, cost, and privacy.

Weaknesses:

- Mainly theoretical in nature.
- Limited practical validation in real deployment scenarios.

Comparison with SPDS:

SPDS translates the general insights of such review studies into a practical System by selecting seat-pressure sensing as a low-cost and privacy-preserving solution. Instead of using a hybrid or complex detection framework, SPDS focuses on a simple implementation that directly addresses the school-bus safety problem.

2.2.4 Study D: Passenger Detection and Counting for Public Transport System

Reference: Passenger Detection and Counting for Public Transport System (ResearchGate, 2020)

This study used computer vision techniques to detect and count passengers entering and leaving public-transport vehicles. The system aimed to improve passenger-tracking accuracy in crowded transport environments.

Strengths:

- Good tracking capability under controlled conditions.
- Useful in moderate crowd-density environments.
- Supports movement-based passenger analysis.

Weaknesses:

- Requires relatively high computational resources.
- Highly dependent on lighting conditions and camera quality.
- Not ideal for privacy-sensitive environments such as school transportation.

Comparison with SPDS: Unlike vision-based systems, SPDS does not rely on image processing or camera placement. This allows SPDS to operate without privacy concerns, reduces hardware complexity, and improves suitability for small-scale and low-cost implementation in school buses.

2.2.5 Study E: Methodology for Evaluating Passenger Counting Technologies

Reference: A Methodology for Empirically Evaluating Passenger Counting Technologies (Swinburne University of Technology, ATRF, 2019)

This study proposed an empirical framework for evaluating passenger-counting systems under real-world operating conditions. It focused on metrics such as accuracy, failure rates, and environmental sensitivity.

Strengths:

- Emphasizes real-world testing and validation.
- Provides useful benchmarking criteria.
- Supports structured performance evaluation.

Weaknesses:

- Focused mainly on transport systems in non-local environments.
- More concerned with evaluation methodology than System simplicity.

Comparison with SPDS:

The evaluation principles presented in this study are useful for SPDS because they provide a basis for testing reliability, responsiveness, and detection accuracy. Such an evaluation approach can help assess SPDS under local operating conditions and improve confidence in the System's performance.

2.2.6 Study F: Wireless Sensor Network Architecture for Passenger Counting

Reference: Wireless Sensor Network Architecture for Passenger Counting (ScienceDirect, 2023)

This study proposed a wireless sensor network (WSN) architecture for distributed passenger counting using low-cost sensor nodes communicating with a central controller. The system emphasized scalability and low power consumption.

Strengths:

- Scalable and low-cost architecture.
- Supports distributed monitoring.
- Suitable for broader transport network applications.

Weaknesses:

- Installation and network management can be complex.
- Accuracy may decrease in crowded conditions.
- More complex than needed for a simple local school-bus System.

Comparison with SPDS:

SPDS follows a similar low-cost sensing idea but avoids the added complexity of wireless networking. Instead of using a full WSN architecture, SPDS relies on direct local sensing and local alerts, which makes it easier to implement, maintain, and test in a simple school-bus safety System.

2.3 Comparison with Related Works (Aligned with Guidelines)

Feature / System	Study A Deep Learning + Camera	Study B FSR + IoT Sensors	Study C Multi-Tech Review	Study D Vision System (Computer Vision)	Study E Empirical Evaluation Framework	Study F Wireless Sensor Network (WSN)
Platform / Type	Camera-based AI system	Seat pressure sensors + IoT module	Hybrid comparative framework	Computer vision detection system	Experimental field model	Wireless IoT network system
Main Features	Object detection and image analysis	Seat occupancy sensing, alerts	Integration of various sensing methods	seat pressure and object tracking	Data collection and validation	Real-time distributed sensing
Level of Autonomy	High (fully automated)	Medium (sensor dependent)	Medium (semi-automated framework)	High (automatic visual tracking)	Medium (manual configuration)	High (self-organized)
Suitable Use	Smart transport systems	School bus monitoring	Research and technology evaluation	seat pressure detection	Performance validation studies	Crowd or passenger systems
Response Time	Seconds (processing dependent)	Instant (sensor trigger)	Variable (simulation-based)	Milliseconds to seconds	Moderate	Real-time response
Portability	Low (camera setup required)	High (lightweight and flexible)	Medium (software-based)	Medium	Low (lab setup)	High (wireless and scalable)
Advantages	High accuracy and recognition rate	Low cost, easy to install	Covers multiple sensing technologies	Accurate detection and tracking	Validated through experiments	Scalable and cost-efficient
Limitations	High cost, privacy concerns	Detects seated passengers only	No real-world testing	Lighting sensitivity	Limited scope	Crowded area interference
Best Users / Contexts	Urban transportation authorities	School and public buses	Research organizations	Surveillance systems	Academic institutions	IoT mobility management
Brief Analysis	Simple, low-cost seat-pressure alternative.	Single seat pressure sensor for occupancy.	Uses one seat sensor, not hybrid.	No lighting issues; pressure-based.	Can follow similar field testing.	Works without multi-sensor networks.

Table 1: Comparison with Related Works

2.4 Identified Gaps and SPDS Contributions

Despite considerable research, key gaps remain:

- Most systems detect either seat occupancy or total passengers, not both.
- Camera-based solutions raise privacy and cost concerns.
- Field testing in Middle Eastern environments is scarce.
- Scalable design that supports future IoT integration for real-time alerts and remote monitoring

SPDS addresses these gaps by offering:

- A sensor-based design detecting seated passengers.
- A non-camera solution preserving privacy and lowering cost.
- Integration with IoT for real-time alerts and remote monitoring.
- Scalability suitable for school and public-bus fleets in Saudi Arabia.

2.5 Conclusion

This chapter reviewed several related studies on passenger detection, passenger counting, seat-occupancy monitoring, and transportation safety systems. The reviewed approaches included deep learning, IoT-based monitoring, computer vision methods, wireless sensor networks, and evaluation frameworks. Although these studies provided useful insights, many of them involved limitations related to privacy, cost, complexity, or limited suitability for school-bus safety applications.

The Smart Passenger Detection System (SPDS) builds on these previous works by adopting a simple pressure-based detection approach that is low-cost, privacy-preserving, and practical for implementation in school buses. The literature review therefore supports the choice of SPDS as a suitable solution for addressing the problem of passengers being left inside buses after the engine is turned off.

The next chapter presents the system analysis and design of SPDS, including the methodology, requirements, and technical structure of the proposed System

System Analysis and Design

3.1 Introduction

This chapter presents the analysis and design of the Smart Passenger Detection System (SPDS). It translates the problem identified in the previous chapters into a clear technical solution that can be implemented as a working system. The main purpose of this phase is to define how the system operates, what components it requires, and how these components interact to improve passenger safety inside school buses.

The proposed system was designed to detect whether a passenger remained on the seat after the engine was turned off. To achieve this, the System relied on a seat pressure sensor, an Arduino Uno microcontroller, an LCD display, a buzzer, an LED indicator, and a control button used to simulate the engine state. Once the engine state changed to OFF, the system checked the seat sensor reading and triggered an alert if seat occupancy was still detected.
This chapter includes the development methodology, system requirements, main system behavior, interface description, and design models used to represent the proposed System.

3.2 Methodology

Several software and system development methodologies are commonly used in embedded-system design, such as the Waterfall model, Agile methodology, and the Spiral model. For the Smart Passenger Detection System (SPDS), the Waterfall model was selected because it is suitable for small-scale embedded systems with clear and fixed requirements.
The Waterfall model follows a sequential process in which each stage is completed before moving to the next one.
This approach was appropriate for SPDS because the project requirements, hardware components, and System objectives were clearly defined from the beginning.
The main stages followed in this project were:

- Requirements analysis and definition
- System design
- Hardware and software implementation
- Testing and evaluation
- Documentation and final reporting

In the context of SPDS:

- During the requirements analysis phase, the main safety problem was identified as the risk of children being forgotten inside buses after the engine was turned off.
- During the design phase, this safety need was translated into a system consisting of a pressure sensor, a controller, a display unit, and audio-visual alert devices.
- During the implementation phase, the hardware components were connected, and the Arduino code was developed to detect seat occupancy and activate alerts.
- During the testing phase, the system behavior was checked under different conditions to verify that the alerts were triggered only when a passenger was detected after the engine state changed to OFF.

3.3 Requirements Analysis and Definition

System requirements define what the system must do and the conditions under which it must operate. In the Smart Passenger Detection System (SPDS), the main objective was to ensure that no passenger remained on the bus seat after the engine was turned off by using automatic detection and alert mechanisms.
To achieve this objective, the system requirements were divided into two categories:

- Functional Requirements: describe the specific operations the system must perform.

- Non-Functional Requirements: describe the quality attributes of the system such as reliability, usability, cost, and safety.

3.3.1 Functional Requirements

A functional requirement defines the specific services or functions that the system must provide.
For the SPDS system, the main actors were:

- Driver: responsible for operating the system and responding to alerts.
- System Controller (Arduino Uno): responsible for reading sensor data, determining seat occupancy, and activating alerts.

A. System Automatic Behavior

The Smart Passenger Detection System performed the following functions automatically:

- Read the engine state using a simulated hardware button.
- Switch to monitoring mode when the engine state changes to OFF.
- Read the seat pressure sensor value continuously.
- Determine seat occupancy by comparing the sensor reading with predefined threshold values.
- Activate an audio-visual alert if the seat remained occupied while the engine stated OFF.
- Display clear status messages on the LCD screen, such as:
- Engine ON
- Seat Empty
- Child Present
- WARNING! Child in Bus

B. Functions Available to the Driver

The system allowed the driver to perform the following actions:

- View the current system status on the LCD screen.
- Receive a sound alert through the buzzer when a passenger was detected after the engine was turned off.
- Observe a visual warning through the LED indicator.
- Change the engine state using a push button in the System to simulate engine ON/OFF conditions.
- Check whether the seat was clear before returning the system to normal operation.

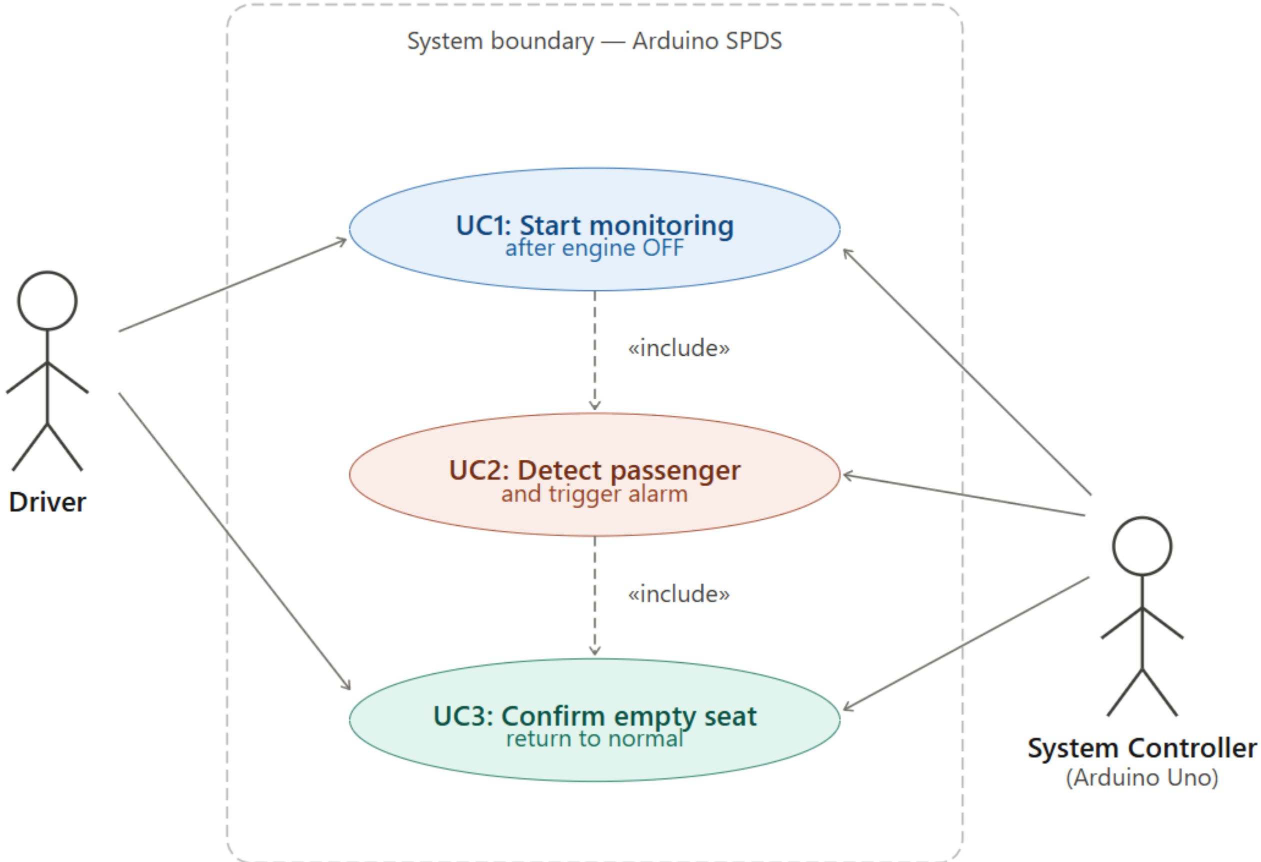

Figure 2: Use case diagram of the Smart Passenger Detection System (SPDS)

3.3.2 Non-Functional Requirements

For SPDS, the main non-functional requirements were:
1. Ease of Use
The system should be simple for the driver to understand. Messages shown on the LCD and the LED indication should make the system state clear without requiring complex interaction.
2. Reliability
The system should consistently detect seat occupancy under normal operating conditions and provide alerts only when needed.
3. Availability
The system should be ready for use whenever the bus is in service and should begin monitoring immediately after the engine state changes to OFF.
4. Safety
The electrical connections should be secure, and the System should not interfere with any primary bus controls in a real deployment scenario.
5. Privacy
The system should not collect personal data, images, or video. This requirement was satisfied because the System used pressure sensing rather than cameras.
6. Low Cost

The selected components, including Arduino Uno, seat pressure sensor, LCD, buzzer, LED, and buttons, should be affordable and easy to obtain.

7. Maintainability

The system should allow simple replacement or adjustment of components such as the sensor, LED, or buttons.

8. Scalability

Although the implemented system used a single seat pressure sensor, the design should allow future expansion to support multiple seats or remote notification features.

3.4 System Design

The Smart Passenger Detection System (SPDS) was designed as a simple embedded safety System. The system architecture consisted of input components, a processing unit, and output components.

Input Components

- Seat pressure sensor
- Engine-state simulation button

Processing Unit

- Arduino Uno microcontroller

Output Components

- LCD display
- Buzzer
- LED indicator

The design logic of the System can be summarized as follows:

1. The system reads the engine state.
2. If the engine is ON, the system remains in normal mode.
3. If the engine changes to OFF, the system checks the seat pressure sensor.
4. If no pressure is detected, the LCD displays that the seat is empty.
5. If pressure is detected, the system activates the buzzer and LED and displays a warning on the LCD.

This design was selected because it provided a direct and low-cost solution to the safety problem while remaining easy to implement and test.

3.4.1 Main Use Case Design: Bus Monitoring After Engine Off

3.4.1.1 User Interface of the system

The driver interacted with the system through a small LCD screen, a push button used to simulate the engine state, a buzzer, and an LED indicator. The LCD displayed the current operating status of the system, including:

- Engine ON
- Engine OFF

- Seat Empty
- Child Present
- WARNING! Child in Bus

The LED acted as a visual warning indicator, while the buzzer provided an audible alarm when a passenger was still detected after the engine was turned off.

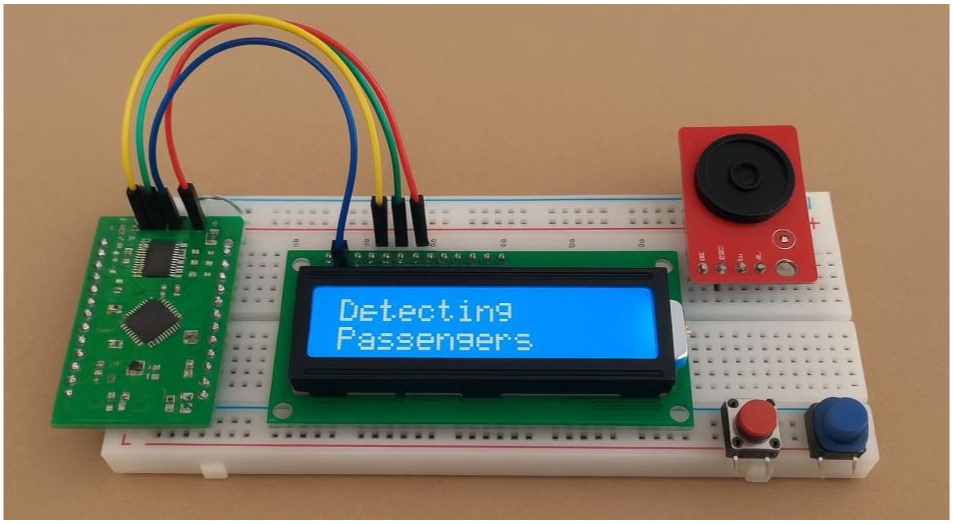

Figure 3: LCD display showing passenger detection status in the System

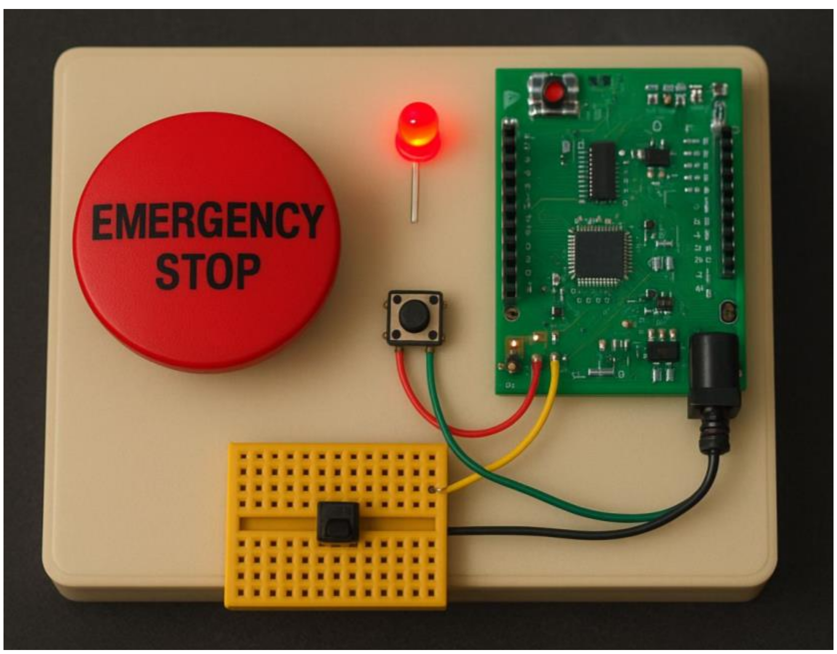

Figure 4: Safety module including the engine button, LED indicator, and buzzer

3.4.1.2 Description of the Main Scenarios

Use Case 1: Start Monitoring After Engine OFF

Abstract:

This use case describes how the system switches from normal operation to monitoring mode once the engine state is changed to OFF.

Actors:

Driver, System Controller (Arduino Uno)

Precondition:

- The engine state is ON.
- The system is operating normally.
- The seat pressure sensor is connected and functioning.

Postcondition:

- The system begins monitoring the seat sensor.
- The LCD displays the current system status.

Main Scenario:

1. The driver changes the engine state to OFF using the System control button.
2. The Arduino detects the engine-state change.
3. The system enters monitoring mode.
4. The LCD updates the system status.
5. The system starts checking the seat pressure value.

Use Case 2: Detect Passenger and Trigger Alarm

Abstract:

This use case describes how the system detects seat occupancy after the engine is turned off and activates an audio-visual warning.

Actors:

System Controller (Arduino Uno)

Precondition:

- The system is in monitoring mode.
- The engine state is OFF.

Postcondition:

- If pressure is detected on the seat, the system enters alert mode.
- The buzzer and LED are activated.
- A warning message appears on the LCD.

Main Scenario:

1. The system reads the seat pressure sensor value.
2. The sensor reading is compared with the configured threshold values.
3. If the reading indicates occupancy, the system identifies that a passenger is still present.
4. The system activates the buzzer.
5. The LED turns on as a visual warning.
6. The LCD displays a warning message such as WARNING! Child in Bus.
7. The driver notices the alert and checks the seat.

Use Case 3: Confirm Empty Seat and Return to Normal State

Abstract:

This use case describes how the system returns to the normal state after the seat becomes empty.

Actors:

Driver, System Controller (Arduino Uno)

Precondition:

- The system is in monitoring or alert mode.
- The driver has checked the bus seat.

Postcondition:

- The system stops the alert.
- The LCD displays that the seat is empty.

Main Scenario:

1. The passenger leaves the seat or the pressure is removed.
2. The seat pressure reading falls below the configured threshold.
3. The system detects that the seat is empty.
4. The buzzer is deactivated.
5. The LED is turned off.
6. The LCD displays Seat Empty.

3.4.1.3 Class Diagram

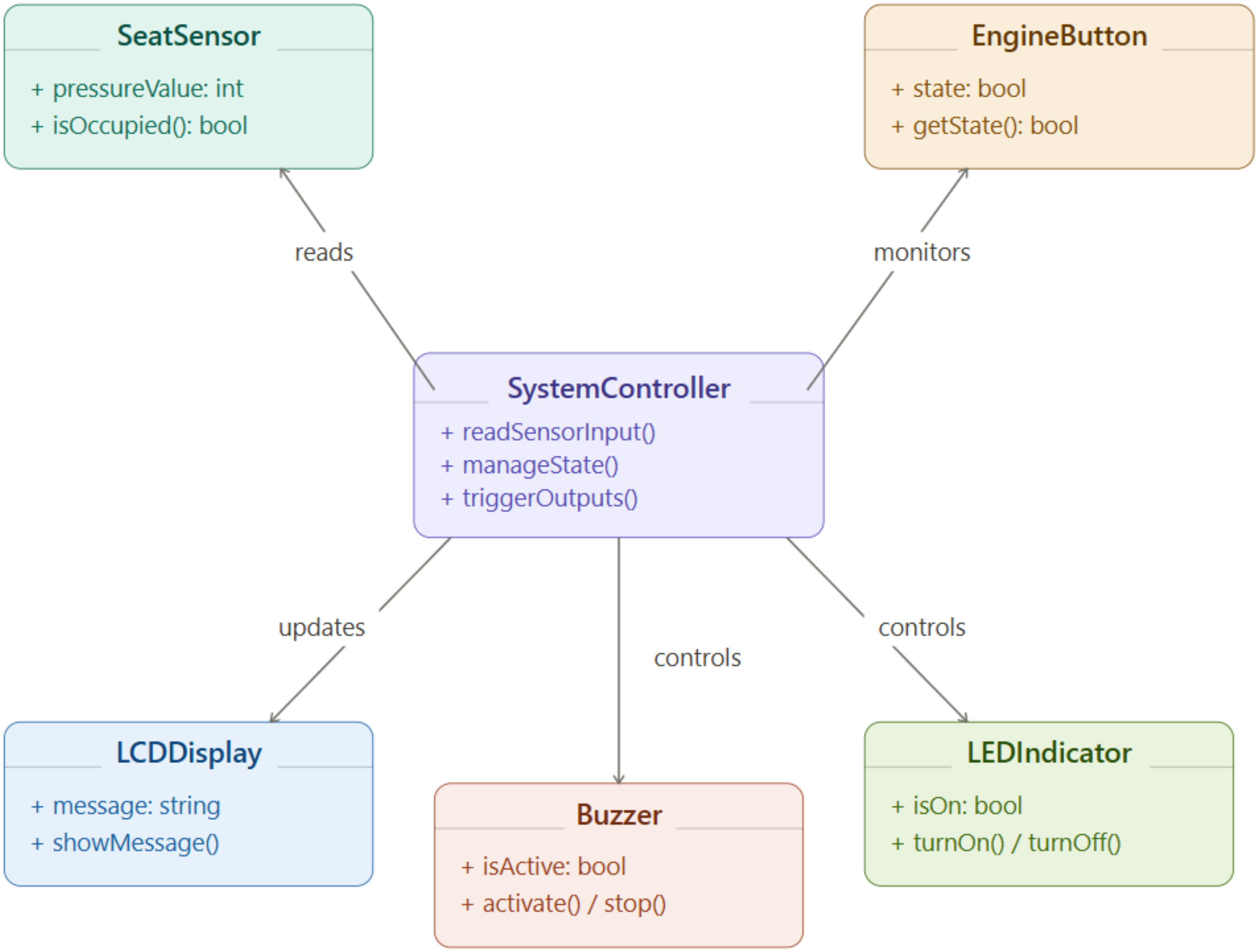

Figure 5: Class diagram of the Smart Passenger Detection System (SPDS)

3.4.1.4 Sequence Diagram

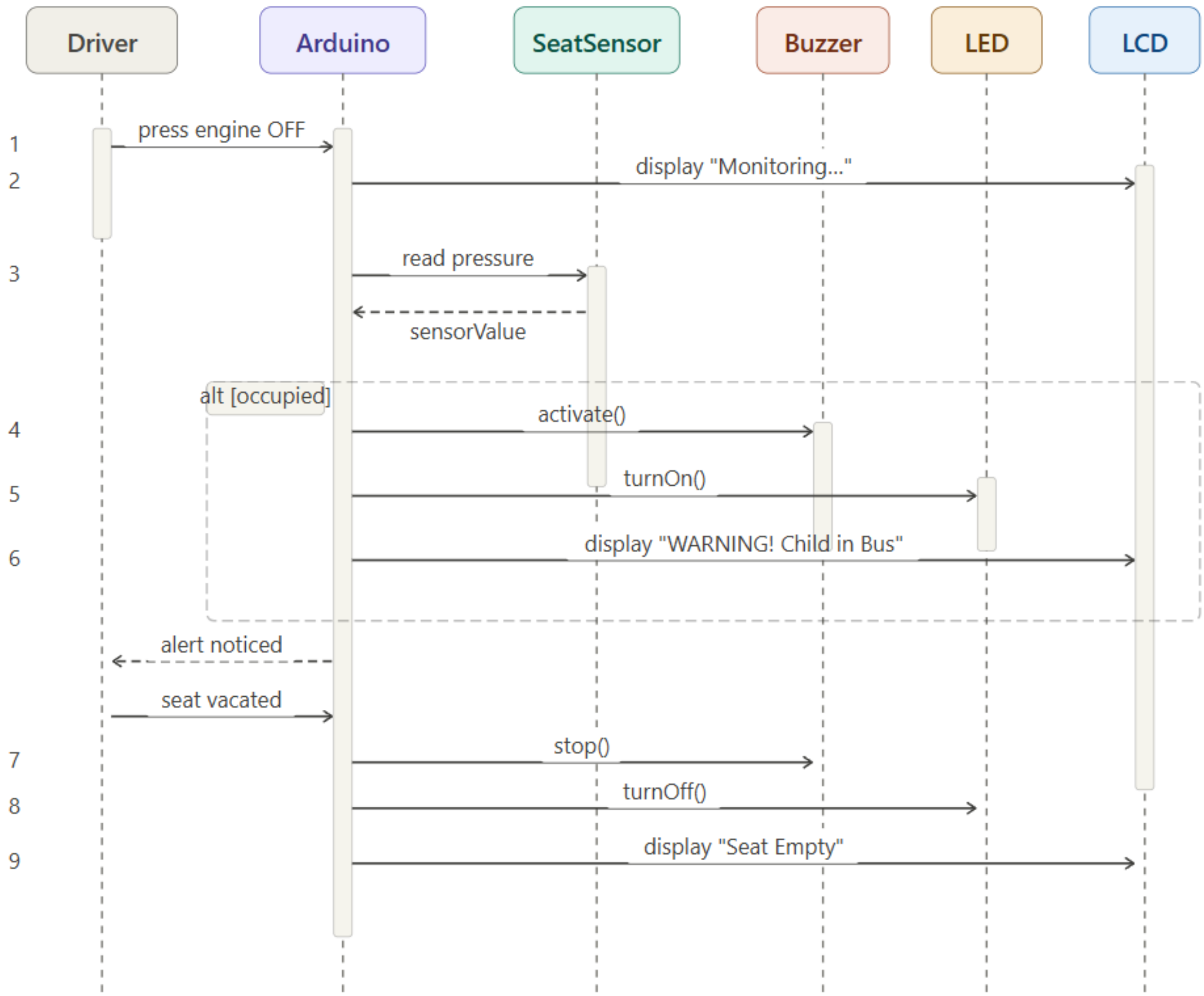

Figure 6: Sequence Diagram

3.5 Logical Structure of the system

The implemented SPDS system did not use a database or data storage module. Instead, the system relied on direct processing of sensor readings through the Arduino Uno microcontroller. The logical structure of the system consisted of the main controller, the seat pressure sensor, the LCD display, the buzzer, the LED indicator, and the engine-state button. These components interacted in real time to detect seat occupancy and activate warning alerts when needed.

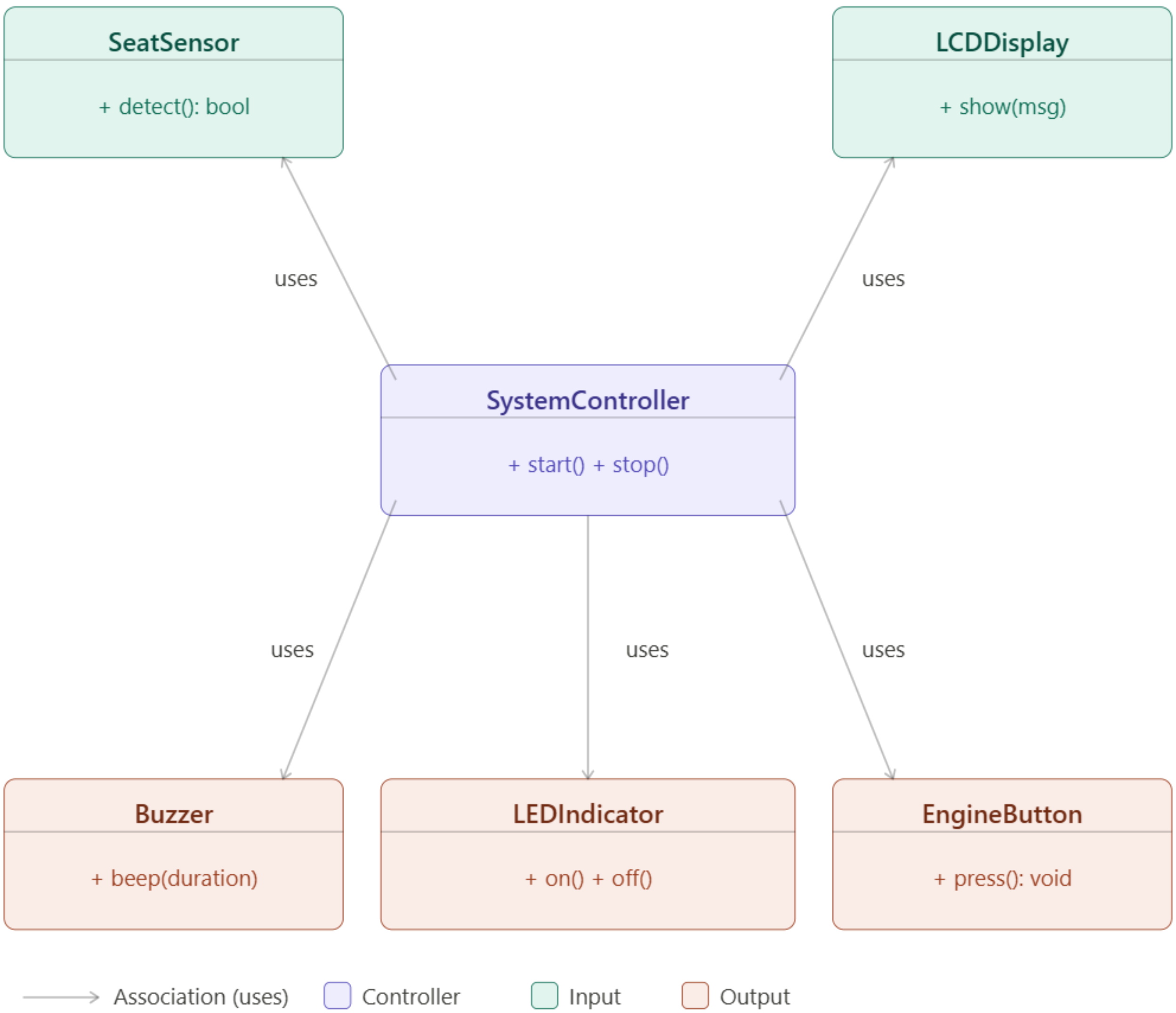

Figure 7: presents the simplified class structure of the implemented system

3.6 Conclusion

This chapter presented the analysis and design of the Smart Passenger Detection System (SPDS). It explained the selected Waterfall development methodology, the functional and non-functional requirements, and the technical design of the System. The chapter also described the system behavior, interface elements, main use cases, and interaction between the system components.

The proposed design showed that a simple pressure-based detection system can provide a practical, privacy-preserving, and low-cost solution for school-bus safety. The next chapter presents the implementation details, hardware and software specifications, testing process, and System results.

Implementation and Tests

4.1 Introduction

This chapter presents the implementation and testing of the Smart Passenger Detection System (SPDS) System. After completing the system analysis and design in the previous chapter, the proposed solution was physically implemented using low-cost hardware components and Arduino-based control logic.

The purpose of this chapter is to describe how the System was built, how the hardware and software components were integrated, and how the system was tested under different operating conditions. The implemented System focused on detecting seat occupancy after the engine state changed to OFF and then activating local audio-visual alerts through an LCD display, a buzzer, and an LED indicator.

This chapter also explains the main testing scenarios, sensor calibration process, and observations made during system evaluation.

4.2 Hardware and Software Specifications

The Smart Passenger Detection System (SPDS) was implemented as a simple embedded System that combined low-cost hardware components with Arduino-based software logic. The System was designed to provide a practical and privacy-preserving school-bus safety solution.

4.2.1 Hardware Specifications

The hardware components used in the implemented System were:

- Arduino Uno Microcontroller

The Arduino Uno was used as the main control unit of the system. It received sensor readings, processed the control logic, and activated the output devices based on the system state.

- Seat Pressure Sensor

A pressure sensor was used to detect whether weight was present on the seat. The sensor reading was treated as the main indicator of passenger presence.

- 16×2 LCD Display (Parallel Interface)

A standard 16×2 LCD was used to display system messages such as engine state, seat status, and warning messages. The display was connected using a parallel interface rather than I2C.

- Buzzer

A buzzer was used to provide an audible warning when the seat remained occupied while the engine state was OFF.

- LED Indicator

An LED was used as a visual warning signal during alert conditions.

- Push Button

A push button was used in the System to simulate the engine ON/OFF state.

- Potentiometer

A potentiometer was used to adjust the LCD contrast.

- Breadboard and Jumper Wires

These were used to connect the components and build the System circuit during implementation and testing.

- Resistors

Resistors were used where needed, such as with the LED and sensor circuit.

4.2.2 Software Specifications

The software environment used in the implementation included:

- Arduino IDE

The Arduino Integrated Development Environment (IDE) was used to write, compile, and upload the source code to the Arduino Uno.

- Arduino C/C++ Programming Language

The system logic was implemented using Arduino C/C++.

- LiquidCrystal Library

The LiquidCrystal library was used to control the 16×2 LCD display.

The software handled:

- seat sensor reading
- threshold comparison
- engine-state control
- LCD status display
- buzzer activation
- LED warning control

The code was also refined to reduce sensor fluctuation by using threshold ranges and averaging logic during testing.

4.3 Implementation

The System was implemented by connecting all hardware components to the Arduino Uno and writing the control program that coordinated system behavior.

The system operated as follows:

1. The push button was used to simulate the engine state.
2. When the engine state was ON, the system displayed the normal operating state.
3. When the engine state changed to OFF, the system began monitoring the seat pressure sensor.
4. If the sensor detected no pressure, the LCD displayed that the seat was empty.
5. If the sensor detected pressure above the configured threshold, the system identified that a passenger was still present.
6. In the alert state, the buzzer was activated, the LED turned on, and the LCD displayed a warning message.

The implemented System displayed messages such as:

- Engine ON
- Engine OFF
- Seat Empty
- Child Present
- WARNING! Child in Bus

During implementation, the seat pressure sensor required calibration because its analog reading did not remain at zero when no weight was applied. Therefore, threshold values were adjusted experimentally until the system could better distinguish between an empty seat and an occupied seat.

The final implementation used two threshold values:

- thresholdHigh = 800
- thresholdLow = 500

These values improved system stability by reducing false detection caused by fluctuating sensor readings during operation.

4.4 Tests

Testing was an essential part of verifying the functionality of the Smart Passenger Detection System. The implemented System was tested to confirm that each component worked correctly on its own and that the integrated system behaved as expected.

4.4.1 Testing Approach

The following testing strategies were used:

- Functional Testing

Each component was tested individually, including the LCD, seat pressure sensor, buzzer, LED, and push button.

- Integration Testing

After individual testing, the components were connected and tested as one integrated System.

- Calibration Testing

Sensor readings were observed under no-load and occupied conditions to determine suitable threshold values.

- Behavior Testing

The system was tested under different engine-state and seat-occupancy conditions to verify correct alert behavior.

4.4.2 Test Scenarios

The following scenarios were used to evaluate the System:

Test 1: LCD Display Test

The LCD was tested to verify that messages were displayed correctly after uploading the code and adjusting the contrast using the potentiometer.

Expected Result:

The LCD displays the correct text according to the system state.

Observed Result:

The LCD successfully displayed system messages after correcting the wiring and contrast adjustment.

Test 2: Seat Pressure Detection Test

The pressure sensor was tested under two conditions:

- no weight on the seat
- weight applied on the seat

The analog readings were monitored and compared to threshold values.

Expected Result:

The system should detect the difference between an empty seat and an occupied seat.

Observed Result:

The sensor produced measurable analog values in both cases, but the readings fluctuated. After calibration, threshold values of 800 and 500 were selected to improve detection stability.

Test 3: Engine OFF Monitoring Test

The push button was used to simulate the engine state. When the engine changed to OFF, the system entered monitoring mode and began checking seat occupancy.

Expected Result:

The system should begin checking the seat condition once the engine is OFF.

Observed Result:

The System correctly switched to monitoring behavior after the engine state changed to OFF.

Test 4: Alarm Activation Test

The seat was kept occupied while the engine state was OFF.

Expected Result:

The LCD should display a warning, the LED should turn on, and the buzzer should activate.

Observed Result:

The system successfully triggered the buzzer and LED and displayed a warning message on the LCD when seat occupancy was detected after engine OFF.

Test 5: Safe State Test

The seat pressure was removed while the engine was OFF.

Expected Result:

The alert should stop and the LCD should indicate that the seat is empty.

Observed Result:

The buzzer stopped, the LED turned off, and the LCD returned to the safe or empty-seat message when occupancy was no longer detected.

4.5 Results and Discussion

The System implementation showed that pressure-based seat monitoring could be used as a practical method for detecting the presence of a passenger after the engine state changed to OFF. The combination of the seat pressure sensor, Arduino Uno, LCD display, buzzer, and LED provided a simple but effective local warning system.

The results showed that:

- the LCD could successfully display the current operating status of the system

- the push button could be used to simulate engine ON/OFF behavior
- the buzzer and LED provided clear warning signals during alert conditions
- the seat pressure sensor could distinguish between occupied and empty states after calibration

However, testing also showed that the pressure sensor did not produce perfectly stable values. Even when no weight was applied, the analog reading was not always zero, and some fluctuation occurred. For this reason, calibration was necessary. The use of upper and lower threshold values improved system reliability and reduced false switching between states.

The implemented system therefore confirmed that a low-cost, privacy-preserving, and hardware-simple solution can support school-bus safety applications. Although the current version was a small-scale system using one seat sensor and local alerts only, the design can be extended in future work to support multiple seats, remote notification, or IoT-based reporting.

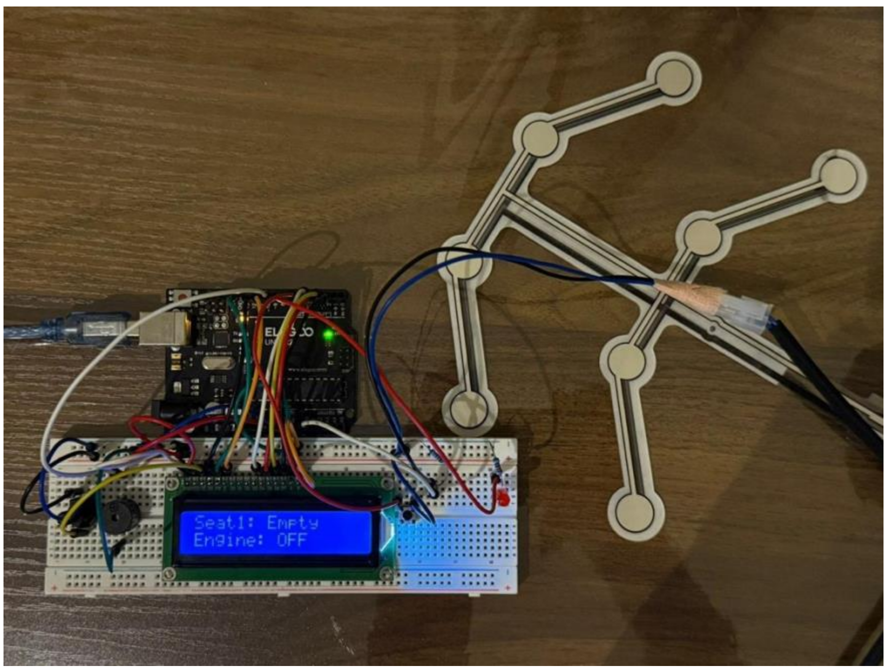

Figure 8: System Overview

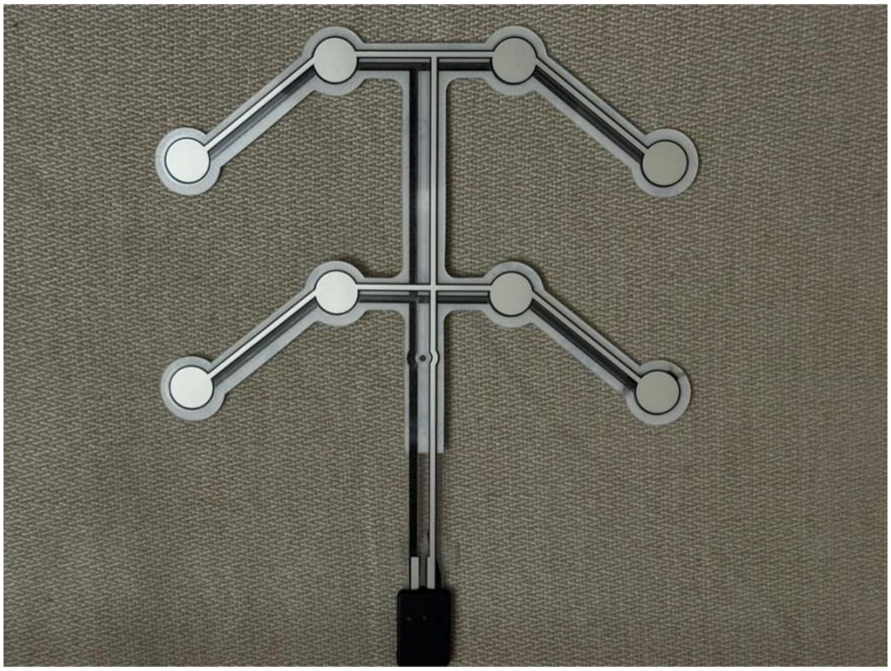

Figure 9: Sensor Placement

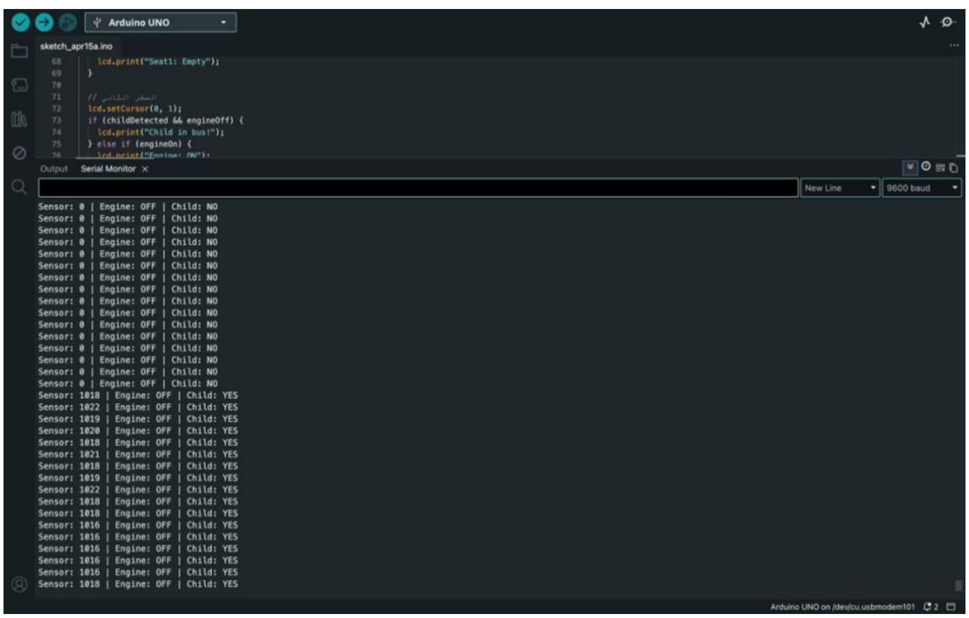

Figure 10: Sensor Readings

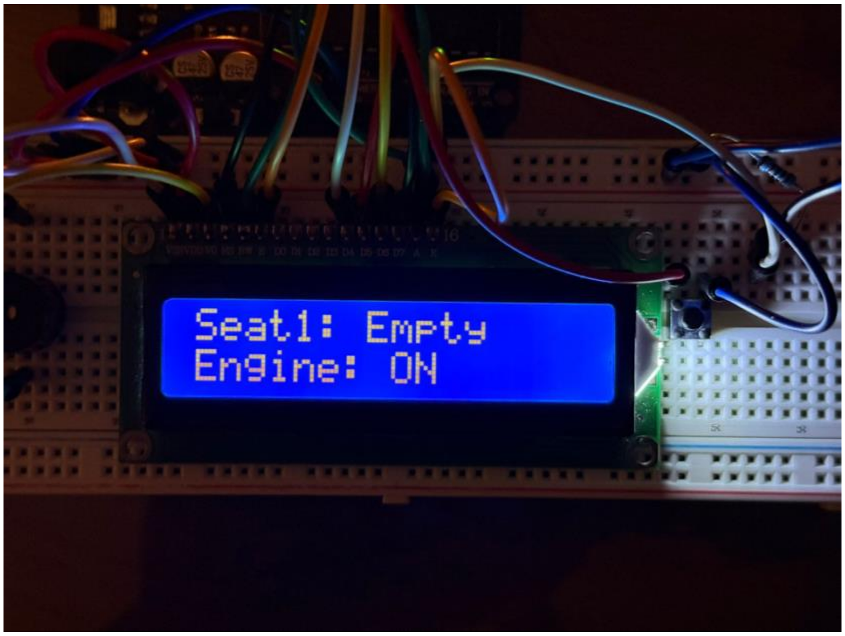

Figure 11: Engine ON Mode

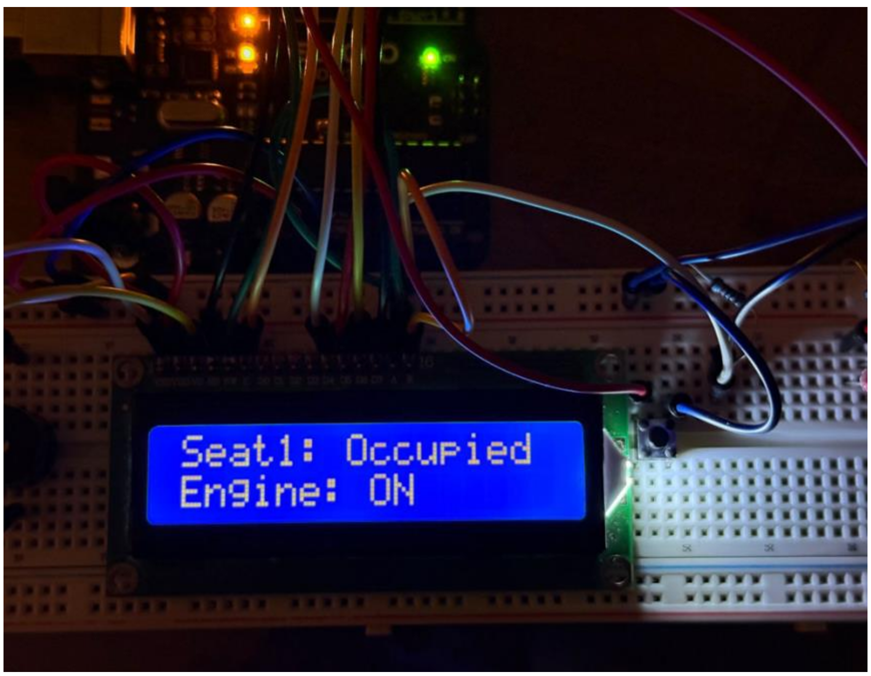

Figure 12: Seat Status Engine On

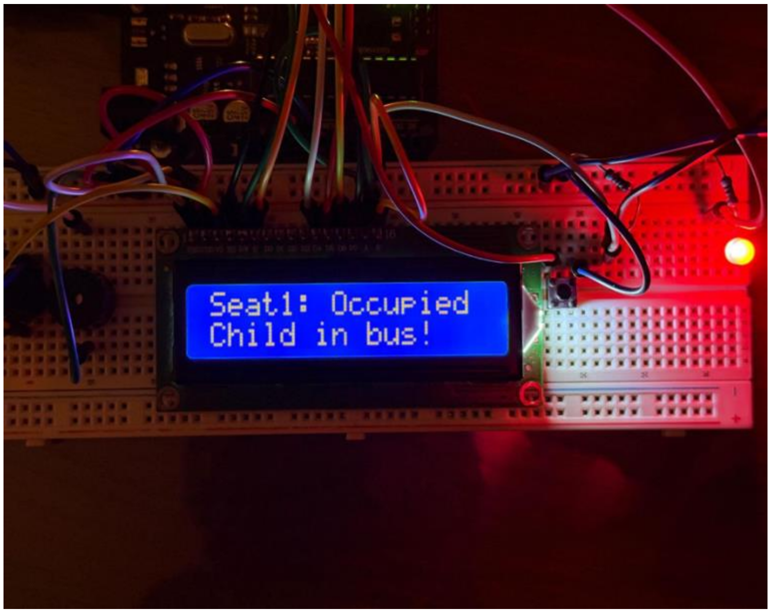

Figure 13: Passenger Detection Alert State

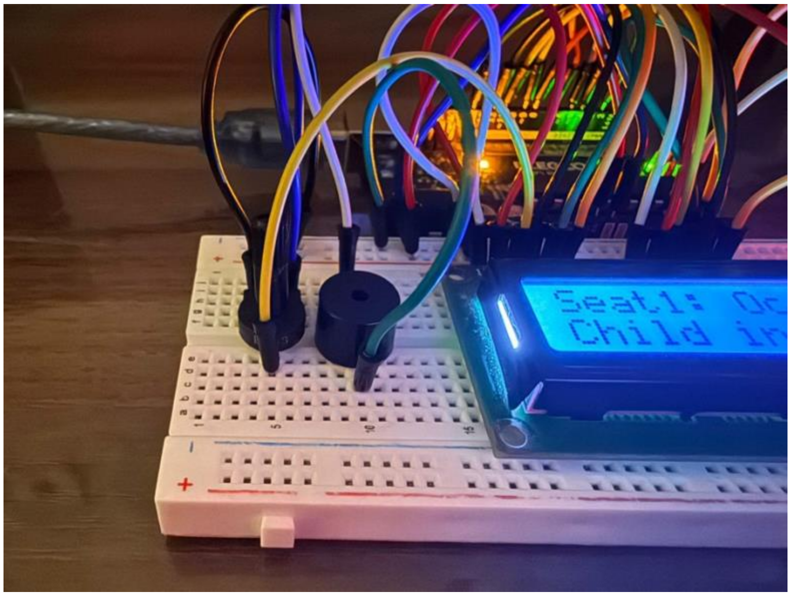
Figure 14: Buzzer Activation Alert

4.6 Conclusion

This chapter presented the implementation and testing of the Smart Passenger Detection System (SPDS) System. It described the hardware and software components used, explained how the System was assembled, and discussed the system behavior during operation.

The testing process showed that the System could successfully monitor seat occupancy after the engine changed to OFF and activate local audio-visual warnings when a passenger was still detected. The LCD, buzzer, LED, and pressure sensor worked together to provide a practical safety mechanism. Although sensor calibration was necessary to improve reliability, the System demonstrated that the proposed system could provide an effective and low-cost school-bus safety solution.

The next chapter presents the final conclusions of the project and suggests possible future enhancements.

Conclusion and Future Works

5.1 Conclusions

The Smart Passenger Detection System (SPDS) project successfully developed and tested a system designed to detect whether a passenger remained on a bus seat after the engine state was turned OFF. The project addressed a critical school-bus safety problem and provided a simple, low-cost, and privacy-preserving solution based on seat-pressure sensing and local audio-visual alerts.

The implemented system used an Arduino Uno, a seat pressure sensor, an LCD display, a buzzer, an LED, and a push button to simulate the engine state. The system was able to monitor the seat condition, detect occupancy after engine OFF, and activate warning signals when needed. The results showed that the system could provide immediate local feedback and improve transportation safety in a practical way.

Although sensor calibration was necessary to reduce fluctuation and improve detection accuracy, the overall system performed successfully and achieved the main project objectives. The system demonstrated that pressure-based sensing can provide an effective alternative to more complex or privacy-sensitive solutions such as camera-based systems.

In conclusion, SPDS provided a practical system for improving school-bus safety and reducing the risk of passengers being forgotten inside buses.

5.2 Recommendations and Future Work

To improve the system, it is recommended to use more stable pressure sensors, improve sensor installation, organize the wiring more effectively, and design a stronger final enclosure for real deployment.

In future work, the system can be extended to support multiple seats, remote notifications, IoT connectivity, mobile applications, data logging, and real-bus testing. These improvements would help transform the System into a more complete and scalable transportation safety system.

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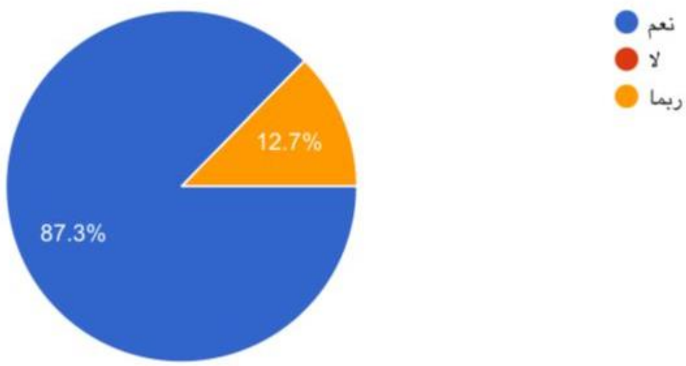
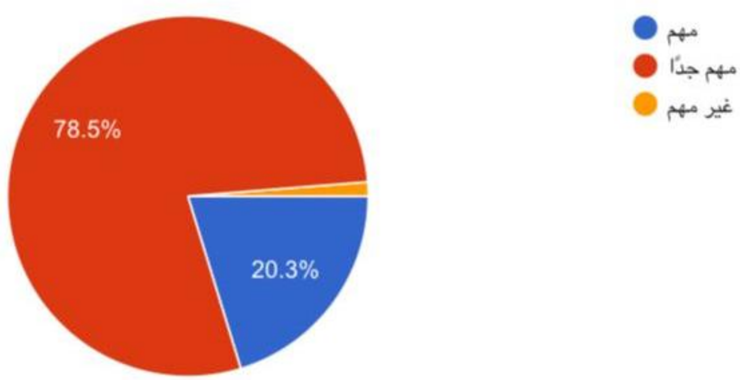
Appendix A

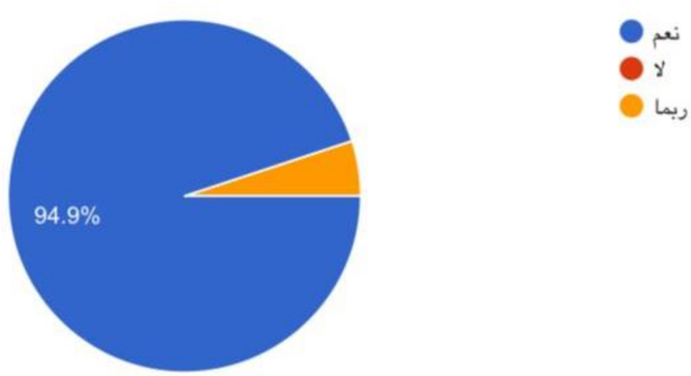
Questionnaire

1. برأيك، هل تُعد مشكلة نسيان الطلاب داخل الحافلات المدرسية مشكلة تحتاج إلى حل تقني؟

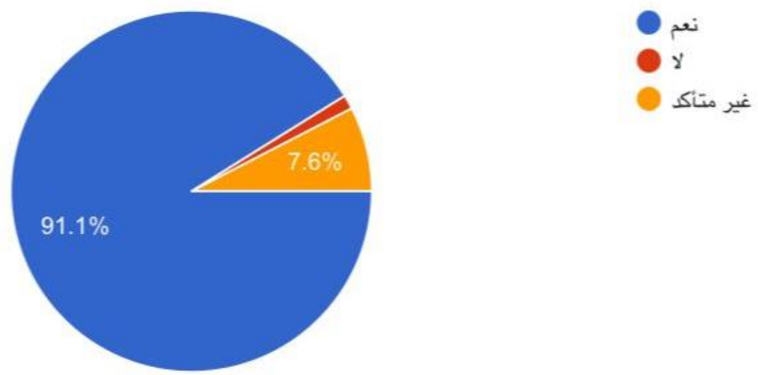


2. ما مدى أهمية وجود نظام ذكي يتحقق تلقائياً من خلو الحافلة من الطلاب؟

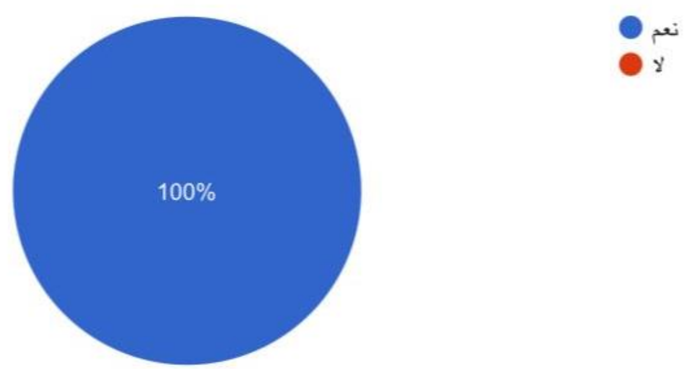




4. **هل تعتقد ان وجود نظام ينبه السائق في حال بقاء طفل داخل الحافلة يمكن ان يساهم في منع الحوادث ؟**



5. **هل تؤيد تطبيق مثل هذا النظام في جميع الحافلات المدرسية لتحسين السلامة؟**



Questionnaire Validation

1	طابع زمني	...يان الطلاب داخل الحافلات المدرسية مشكلة تحتاج إلى حل تقني ؟	...ى أهمية وجود نظام ذكي يتحقق تلقائياً من خلو الحافلة من الطلاب؟	تقليل الحوادث الناتجة عن نسيان الطلاب داخل الحافلات المدرسية ؟	في حال بقاء طفل داخل الحافلة يمكن أن يساهم في منع الحوادث ؟	...طبيق مثل هذا النظام في جميع الحافلات المدرسية لتحسين السلامة؟	...اعد في تحسين فكرة نظام الكشف الذكي داخل الحافلات المدرسية؟
2	12:43:23 2025/12/08 ص	نعم	مهم	نعم	غير متأكد	نعم	مشروع خيا!!!!لي و فكرة خيالية و تقلل من نسيان الطلاب في الحاف
3	12:44:29 2025/12/08 ص	نعم	مهم جداً	نعم	نعم	نعم	الله يوفقنا وإياكم يارب
4	12:47:18 2025/12/08 ص	نعم	مهم جداً	نعم	نعم	نعم	فكره رائعه لا توجد اي ملاحظات
5	12:52:26 2025/12/08 ص	نعم	مهم	نعم	نعم	نعم	مشروعكم افضل مشروع مايحتاج اطرح نوتس
6	1:47:23 2025/12/08 ص	ربما	مهم	نعم	نعم	نعم	ولا شي
7	2:22:56 2025/12/09 ص	نعم	مهم جداً	نعم	نعم	نعم	طبقو الفكره الان
8	3:11:47 2025/12/09 ص	نعم	مهم جداً	نعم	نعم	نعم	النظام التقني مهم جداً لسلامة ركاب الحافلة
9	5:03:18 2025/12/09 ص	نعم	مهم جداً	نعم	نعم	نعم	لا
10	6:32:13 2025/12/09 ص	نعم	مهم جداً	نعم	نعم	نعم	نظام تنبيه متعدد المستويات يقترح أن تكون الإنذارات على 3 مراحل
11	6:59:38 2025/12/09 ص	نعم	مهم	نعم	نعم	نعم	لا ليس هناك ملاحظات
12	9:53:35 2025/12/09 ص	نعم	مهم جداً	نعم	نعم	نعم	لا
13	12:47:32 2025/12/09 م	نعم	مهم جداً	نعم	نعم	نعم	لا
14	12:49:02 2025/12/09 م	نعم	مهم جداً	ربما	نعم	نعم	الطفل امانة عند المدرسة والوزارة الى ان يرجع الى المنزل
15	6:00:09 2025/12/09 م	نعم	مهم جداً	نعم	نعم	نعم	فكرة رائعة تستحق الشكر والاشادة إذ إنّ توظيف التقنية في خدمة ا
16	6:03:31 2025/12/09 م	نعم	مهم جداً	نعم	نعم	نعم	لا
17	6:06:41 2025/12/09 م	نعم	مهم جداً	نعم	نعم	نعم	لا
18	6:07:09 2025/12/09 م	نعم	مهم جداً	نعم	نعم	نعم	اتمنى التطبيق السريع على جميع الحافلات
19	6:09:04 2025/12/09 م	نعم	مهم جداً	نعم	غير متأكد	نعم	ركوب الطلاب ببطاقات تسجل الدخول والخروج من الحافلة - وجود ج
20	6:11:51 2025/12/09 م	نعم	مهم جداً	نعم	نعم	نعم	-
21	6:13:08 2025/12/09 م	نعم	مهم	نعم	نعم	نعم	لا
22	6:13:47 2025/12/09 م	نعم	مهم جداً	نعم	نعم	نعم	لايوجد
23	6:14:35 2025/12/09 م	نعم	مهم جداً	نعم	نعم	نعم	لا
24	6:17:02 2025/12/09 م	نعم	مهم	ربما	نعم	نعم	فحص قائد الحافله والتاكد من سلامته فكريا وعقليا لان الامانه مطلو
25	6:17:59 2025/12/09 م	نعم	مهم جداً	نعم	نعم	نعم	-
26	6:18:59 2025/12/09 م	نعم	مهم جداً	نعم	نعم	نعم	نعم , يكون هناك جهاز لدى السائق ولدى الركاب وعند بقاء احد الطلا
27	6:19:30 2025/12/09 م	نعم	مهم جداً	نعم	نعم	نعم	تصنيع الحافلات نفسها مثل مايوجد كميرا لتحقيق سلامة سير الباص
28	6:25:28 2025/12/09 م	نعم	مهم	نعم	نعم	نعم	لا يوجد
29	6:32:37 2025/12/09 م	نعم	مهم جداً	نعم	نعم	نعم	وضع زر تنبيه في آخر الحافلة ويطلب من الطلاب الضغط عليه عند ا
30	6:32:52 2025/12/09 م	نعم	مهم جداً	نعم	نعم	نعم	لا
31	6:35:45 2025/12/09 م	نعم	مهم جداً	نعم	نعم	نعم	لا
32	6:40:42 2025/12/09 م	نعم	مهم جداً	نعم	نعم	نعم	لا توجد
33	6:43:38 2025/12/09 م	ربما	مهم جداً	نعم	نعم	نعم	أود تركيب كاميرات لمتابعه الطالبات كذلك تستطيع مسؤوله النقل المدر
34	6:45:28 2025/12/09 م	نعم	مهم جداً	نعم	نعم	نعم	توعية الطلاب عن أهمية النهوض والانتباه ، التأكد بين فترة وفترة عن
35	6:53:33 2025/12/09 م	نعم	مهم جداً	نعم	نعم	نعم	لا
36	7:12:33 2025/12/09 م	نعم	مهم جداً	نعم	نعم	نعم	.
37	7:22:43 2025/12/09 م	نعم	مهم جداً	نعم	نعم	نعم	لا
38	7:27:12 2025/12/09 م	نعم	مهم جداً	نعم	نعم	نعم	من الضروري وجود مشرف ف الحافلات منع النزاعات والمشاكل
39	7:37:01 2025/12/09 م	نعم	مهم جداً	نعم	نعم	نعم	لا
40	7:47:43 2025/12/09 م	نعم	مهم جداً	ربما	غير متأكد	نعم	لايوجد
41	8:08:05 2025/12/09 م	نعم	مهم جداً	نعم	نعم	نعم	لا
42	8:09:17 2025/12/09 م	نعم	مهم جداً	نعم	نعم	نعم	وجود كاميرا مراقبه داخل الحافله للحد من نسيان الطالبات كذلك الم
43	8:11:15 2025/12/09 م	نعم	مهم	نعم	نعم	نعم	تطبيق ذكي
44	8:17:47 2025/12/09 م	نعم	مهم جداً	نعم	نعم	نعم	فكره جيده أو أيدها وتطبيقها يساهم كثيرا في تقليل الحوادث داخل
45	8:24:37 2025/12/09 م	نعم	مهم جداً	نعم	نعم	نعم	وضع كاميرات مراقبه داخل الحافلات
46	8:26:12 2025/12/09 م	ربما	مهم	ربما	لا	نعم	إنشاء نظام موثوق يحدد وجود الطلاب في الحافله ام لا، ويرسل إلى
47	8:38:50 2025/12/09 م	نعم	مهم	نعم	نعم	نعم	وجود مشرف مع السائق للتاكد من ان جميع الطلاب نزلو
48	8:44:32 2025/12/09 م	نعم	مهم جداً	نعم	نعم	نعم	ممكن إضافة كاميرات ذكية مع حساسات حركة وحرارة للتأكد من عد
49	8:45:43 2025/12/09 م	نعم	مهم جداً	نعم	نعم	نعم	لا
50	8:47:33 2025/12/09 م	نعم	مهم جداً	نعم	نعم	نعم	الدمج مع الأنظمة المدرسية ربط مباشر بالحضور والغياب: دمج بيانا
51	8:58:28 2025/12/09 م	ربما	مهم جداً	نعم	غير متأكد	نعم	ضروري يكون موجود في جميع المدارس 🤝
52	9:01:40 2025/12/09 م	نعم	مهم جداً	نعم	نعم	نعم	يكون التطبيق مرتبط بالاهل والسائق
53	9:03:03 2025/12/09 م	نعم	مهم جداً	نعم	نعم	نعم	لا
54	9:13:09 2025/12/09 م	ربما	مهم	نعم	نعم	نعم	ممكن النظام يكون استشعار للاجسام
55	9:15:32 2025/12/09 م	ربما	مهم جداً	نعم	غير متأكد	نعم	لا
56	9:22:37 2025/12/09 م	نعم	مهم	نعم	نعم	نعم	no
57	9:29:04 2025/12/09 م	نعم	مهم جداً	نعم	نعم	نعم	ربط النظام على مكاتب ادارة الباصات للمراقبه
58	9:41:10 2025/12/09 م	ربما	غير مهم	نعم	نعم	نعم	.
59	10:16:35 2025/12/09 م	نعم	مهم	نعم	نعم	نعم	لا
60	10:44:45 2025/12/09 م	ربما	مهم	نعم	نعم	نعم	ادعم هذا النظام لغالبية حدوث مثل هذه المواقف
61	1:15:26 2025/12/10 ص	نعم	مهم جداً	نعم	نعم	نعم	لا
62	3:52:12 2025/12/10 ص	نعم	مهم جداً	نعم	نعم	نعم	أن يكون نظام بصمة بطاقة الكترونية مع الطالب عند الدخول والخرو
63	7:28:15 2025/12/10 ص	نعم	مهم جداً	نعم	نعم	نعم	لايوجد
64	8:22:35 2025/12/10 ص	نعم	مهم جداً	نعم	نعم	نعم	لا
65	8:41:38 2025/12/10 ص	ربما	مهم	نعم	نعم	نعم	لا
66	12:58:11 2025/12/10 م	ربما	مهم	نعم	غير متأكد	نعم	لا بالتوفيق
67	2:40:41 2025/12/10 م	نعم	مهم جداً	نعم	نعم	نعم	الفكره رائعه
68	2:48:56 2025/12/10 م	نعم	مهم جداً	نعم	نعم	نعم	بالتوفيق
69	2:55:37 2025/12/10 م	نعم	مهم جداً	نعم	نعم	نعم	لا يوجد
70	2:58:57 2025/12/10 م	نعم	مهم جداً	نعم	نعم	نعم	اتمنى تطبيق برنامج ينبه بوجود طفل داخل المراكب أو الباصات
71	3:13:22 2025/12/10 م	نعم	مهم جداً	نعم	نعم	نعم	الافضل يكون مربوط جهاز الانذار على كل مقعد في الحافله في حال
72	3:26:37 2025/12/10 م	نعم	مهم جداً	نعم	نعم	نعم	يمكن وضع بطاقات تعريف لكل الطلاب بحيث كل طالب او طالبه يص
73	4:50:06 2025/12/10 م	نعم	مهم جداً	نعم	نعم	نعم	لايوجد
74	4:54:47 2025/12/10 م	نعم	مهم جداً	نعم	نعم	نعم	لا
75	5:31:06 2025/12/10 م	نعم	مهم جداً	نعم	نعم	نعم	لا
76	9:12:00 2025/12/10 م	نعم	مهم جداً	نعم	نعم	نعم	لابد من وجود مشرفه تتابع الاطفال
77	9:21:05 2025/12/10 م	نعم	مهم جداً	نعم	نعم	نعم	ليس لدي
78	11:30:30 2025/12/10 م	نعم	مهم جداً	نعم	نعم	نعم	لا
79	4:06:59 2025/12/11 ص	نعم	مهم جداً	نعم	نعم	نعم	لا
80	10:05:25 2025/12/11 ص	نعم	مهم جداً	نعم	نعم	نعم	النظام ممتاز بإذن الله تعالى وقوته

Figure 15: Questionnaire Validation

Questionnaire Results

The questionnaire results are shown below

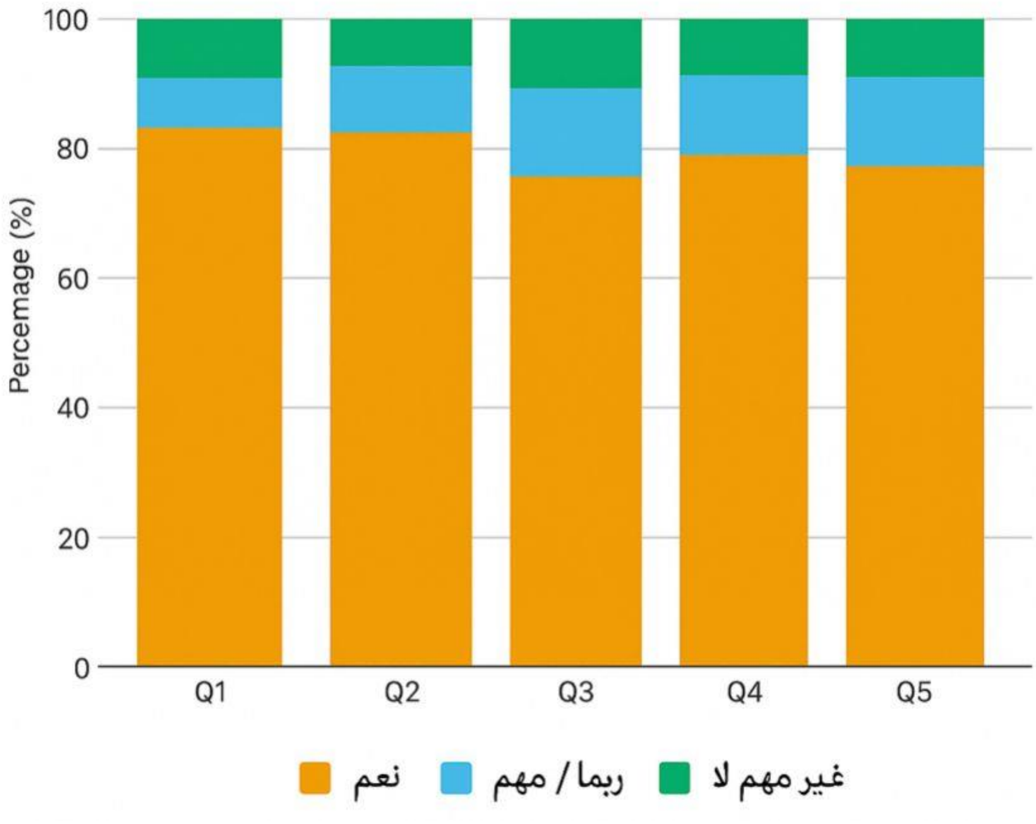


Figure 16: Questionnaire Result